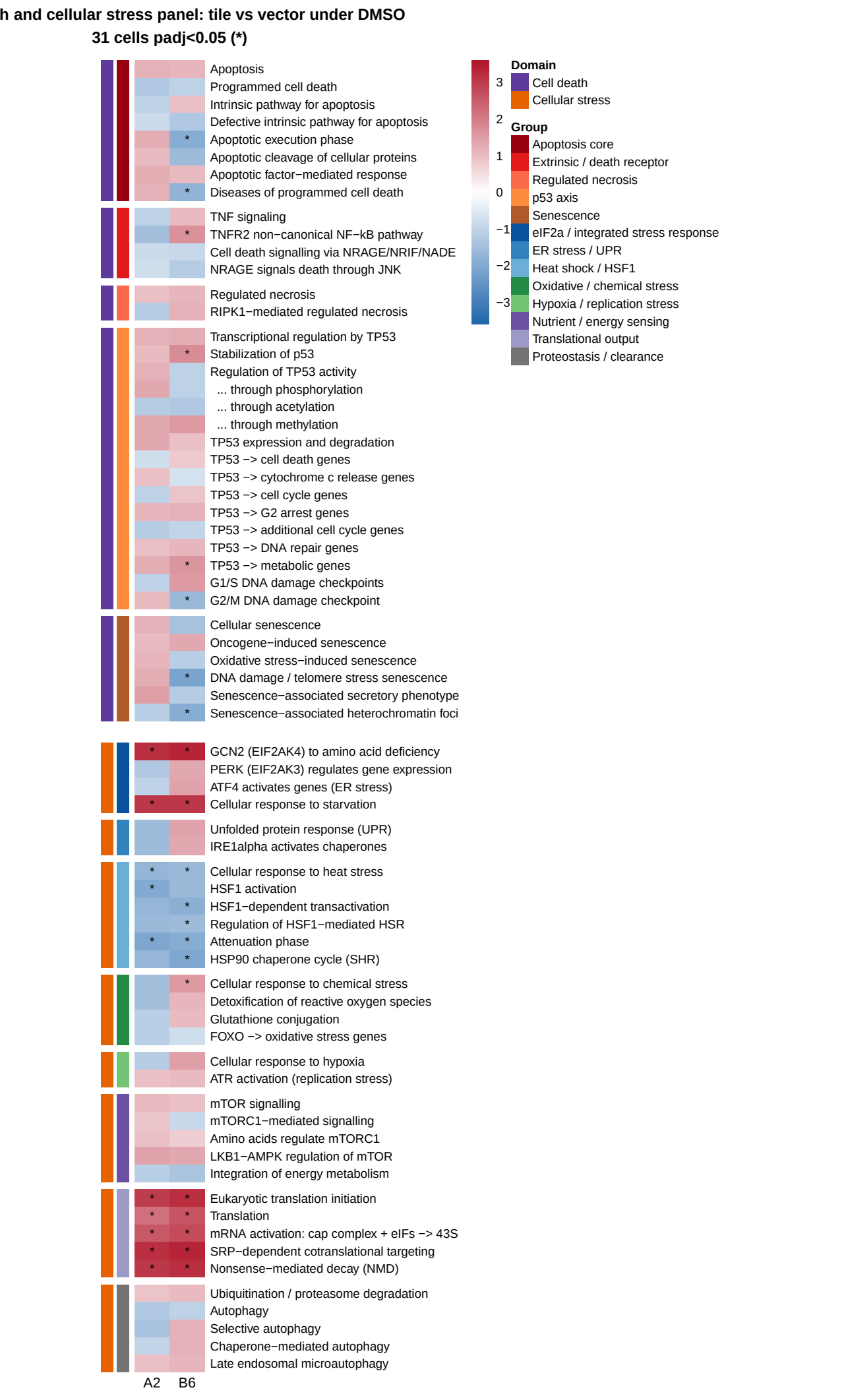
h and cellular stress panel: tile vs vector under DMSO
31 cells padj<0.05 (*)
Apoptosis
Programmed cell death
Intrinsic pathway for apoptosis
Defective intrinsic pathway for apoptosis
Apoptotic execution phase
Apoptotic cleavage of cellular proteins
Apoptotic factor−mediated response
Diseases of programmed cell death
TNF signaling
TNFR2 non−canonical NF−kB pathway
Cell death signalling via NRAGE/NRIF/NADE
NRAGE signals death through JNK
Regulated necrosis
RIPK1−mediated regulated necrosis
Transcriptional regulation by TP53
Stabilization of p53
Regulation of TP53 activity
... through phosphorylation
... through acetylation
... through methylation
TP53 expression and degradation
TP53 −> cell death genes
TP53 −> cytochrome c release genes
TP53 −> cell cycle genes
TP53 −> G2 arrest genes
TP53 −> additional cell cycle genes
TP53 −> DNA repair genes
TP53 −> metabolic genes
G1/S DNA damage checkpoints
G2/M DNA damage checkpoint
Cellular senescence
Oncogene−induced senescence
Oxidative stress−induced senescence
DNA damage / telomere stress senescence
Senescence−associated secretory phenotype
Senescence−associated heterochromatin foci
GCN2 (EIF2AK4) to amino acid deficiency
PERK (EIF2AK3) regulates gene expression
ATF4 activates genes (ER stress)
Cellular response to starvation
Unfolded protein response (UPR)
IRE1alpha activates chaperones
Cellular response to heat stress
HSF1 activation
HSF1−dependent transactivation
Regulation of HSF1−mediated HSR
Attenuation phase
HSP90 chaperone cycle (SHR)
Cellular response to chemical stress
Detoxification of reactive oxygen species
Glutathione conjugation
FOXO −> oxidative stress genes
Cellular response to hypoxia
ATR activation (replication stress)
mTOR signalling
mTORC1−mediated signalling
Amino acids regulate mTORC1
LKB1−AMPK regulation of mTOR
Integration of energy metabolism
Eukaryotic translation initiation
Translation
mRNA activation: cap complex + eIFs −> 43S
SRP−dependent cotranslational targeting
Nonsense−mediated decay (NMD)
Ubiquitination / proteasome degradation
Autophagy
Selective autophagy
Chaperone−mediated autophagy
Late endosomal microautophagy
A2
B6
3
2
1
0
−1
−2
−3
Domain
Cell death
Cellular stress
Group
Apoptosis core
Extrinsic / death receptor
Regulated necrosis
p53 axis
Senescence
eIF2a / integrated stress response
ER stress / UPR
Heat shock / HSF1
Oxidative / chemical stress
Hypoxia / replication stress
Nutrient / energy sensing
Translational output
Proteostasis / clearance